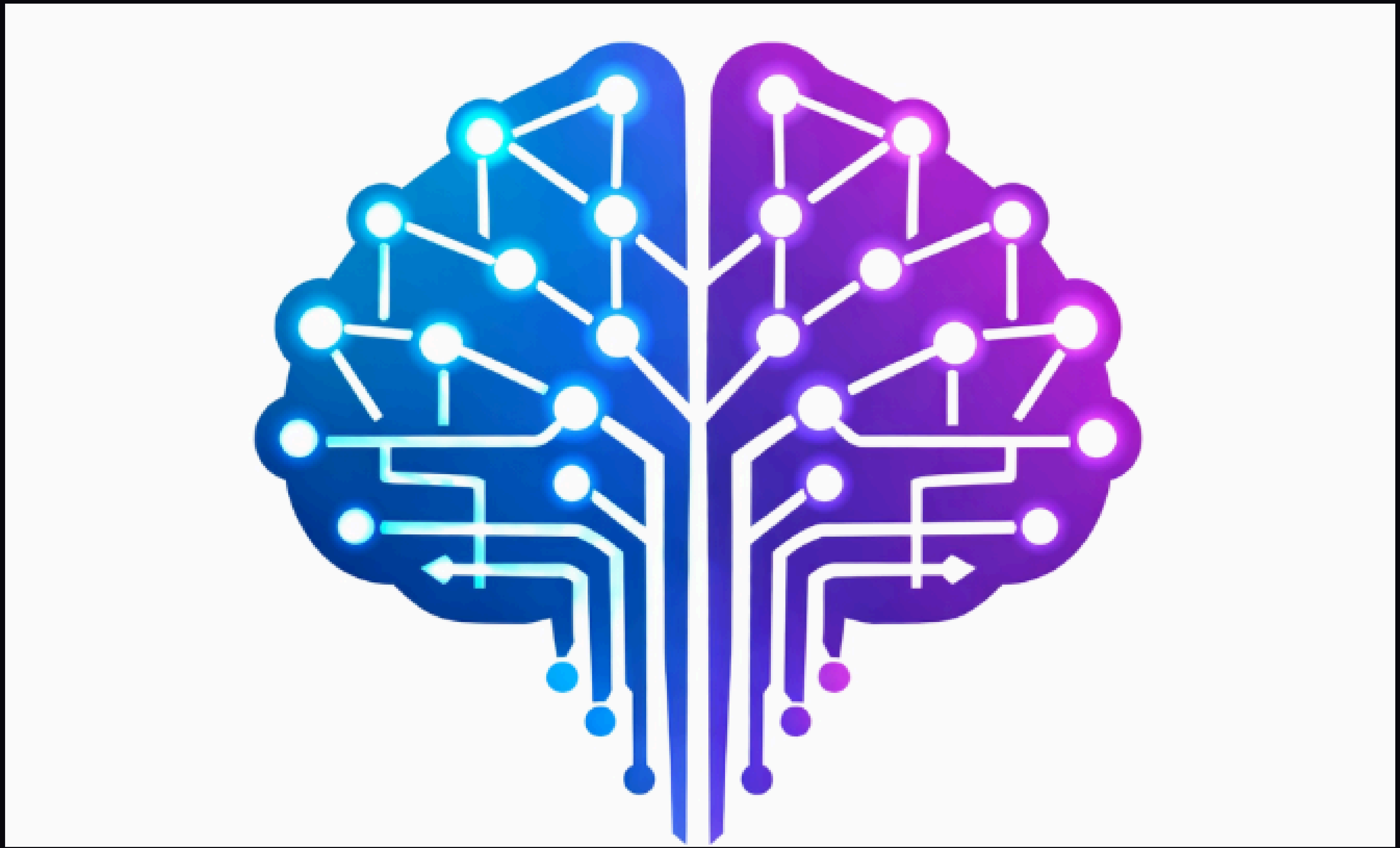
GENAI FUNDAMENTALS



LECTURE:

EMBEDDINGS & SEMANTIC UNDERSTANDING

Why Machines Need a Way to Represent Meaning

- Computers cannot understand language directly.
- To process text, machines convert language into numerical representations.
- These representations are called embeddings.
- Embeddings allow models to capture semantic meaning in numbers.

What Is an Embedding?

An embedding is a vector of numbers representing the meaning of text.

Example (simplified):
"dog" → [0.12, -0.45, 0.88, 0.31, ...]

The vector captures semantic relationships between words and sentences.

Semantic Similarity

Embeddings allow machines to measure meaning similarity.

Example:

Sentence A: “Dogs are great pets.”

Sentence B: “Dogs make wonderful companions.”

These sentences have similar embeddings.

Visualizing Embedding Space

Embeddings exist in a high-dimensional vector space.
Similar concepts appear closer together.

Example clusters:
Animals → cat, dog, horse
Vehicles → car, truck, bus
Programming → Python, Java, C++

Embeddings Enable Semantic Search

Embeddings allow systems to search by *meaning instead of keywords*.

Example:

- Query: "How do I train a neural network?"
- Relevant document: "Steps for training deep learning models."

Semantic similarity helps retrieve relevant information.

Where Embeddings Are Used

Embeddings power many AI applications:

- semantic search
- recommendation systems
- document retrieval
- clustering similar content
- Retrieval-Augmented Generation (RAG)

Key Takeaway

Embeddings convert text into numerical vectors representing meaning.

This enables machines to:

- measure semantic similarity
- retrieve relevant documents
- understand relationships between ideas

Embeddings are a foundation of modern GenAI systems.